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OPERATIONS RESEARCH 415 PART IV: QUEUING THEORY

LECTURE 26: BIRTH & DEATH PROCESSES

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Birth & Death Processes (pp. 1063–1064)

The number of people present at time t in a queuing system is called the **state** of the system at time t .

Laws of Motion of a birth & death process in state j at time t :

- 1 With probability $\lambda_j \Delta t + o(\Delta t)$, a **birth** occurs between time t and time $t + \Delta t$. A birth increases the system state by 1, to $j + 1$. The parameter λ_j is called the **birth rate** in state j . *In most queuing systems, a birth is simply an arrival.*
- 2 With probability $\mu_j \Delta t + o(\Delta t)$, a **death** occurs between time t and time $t + \Delta t$. A death decreases the system state by 1, to $j - 1$. The parameter μ_j is the **death rate** in state j . *In most queuing systems, a death is a service completion.* Note that $\mu_0 = 0$ must hold, or a negative state could occur.
- 3 Births and deaths are **independent** of each other.

Laws 1–3 can be used to show that the probability that more than one event (birth or death) occurs between time t and $t + \Delta t$ is $o(\Delta t)$ and may hence be ignored.

The System $M/M/1/FCFS/\infty/\infty$ (pp. 1064–1065)

Consider an $M/M/1/FCFS/\infty/\infty$ queuing system in which inter-arrival times are exponentially distributed with parameter λ and service times are exponentially distributed with parameter μ .

If the state at time t is j , then the no-memory property of the exponential distribution implies that the probability of a birth during the time interval $[t, t + \Delta t]$ does not depend on how long the system has been in state j and thus may be determined as if an arrival had just occurred at time $t = 0$.

Hence the probability of a birth occurring during $[t, t + \Delta t]$ is

$$\begin{aligned} \int_0^{\Delta t} \lambda e^{-\lambda t} dt &= 1 - e^{-\lambda \Delta t} = 1 - \left(1 - \lambda \Delta t + \overbrace{\frac{(\lambda \Delta t)^2}{2!} - \frac{(\lambda \Delta t)^3}{3!} + \dots}^{o(\Delta t) \text{ by Taylor}} \right) \\ &= \lambda \Delta t + o(\Delta t). \end{aligned}$$

We conclude that the birth rate in state j is simply the arrival rate λ .

The System $M/M/1/FCFS/\infty/\infty$ (pp. 1064–1065)

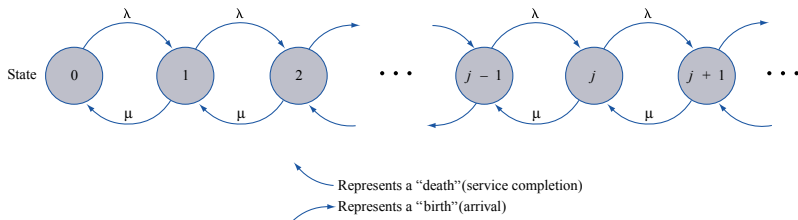
Similarly, the no-memory property of the exponential distribution implies that the probability of a customer completing service between time t and $t + \Delta t$ is given by

$$\int_0^{\Delta t} \mu e^{-\mu \Delta t} dt = 1 - e^{-\mu \Delta t} = \mu \Delta t + o(\Delta t).$$

So the death rate in state $j \geq 1$ is simply the service rate μ .

Hence the $M/M/1/FCFS/\infty/\infty$ queuing system is a birth-death process!

Rate diagram:



The System $M/M/3/FCFS/\infty/\infty$ (p. 1065)

Consider the system $M/M/3/FCFS/\infty/\infty$ in which inter-arrival times are exponentially distributed with $\lambda = 4$ and service times are exponentially distributed with $\mu = 5$.

$$\lambda_j = 4, \quad j = 0, 1, 2, \dots$$

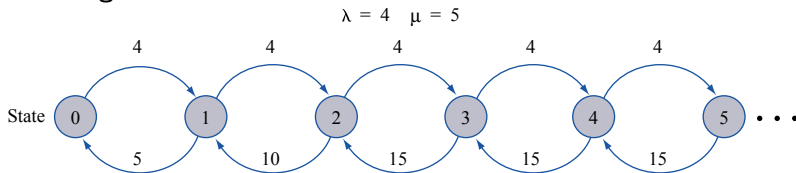
$$\mu_0 = 0,$$

$$\mu_1 = 5,$$

$$\mu_2 = 5 + 5 = 10,$$

$$\mu_j = 5 + 5 + 5 = 15, \quad j = 3, 4, 5, \dots$$

Rate diagram:



The System $M/G/1/FCFS/\infty/\infty$ (p. 1065)

Not all queuing systems can be modelled as birth-death processes!

Consider the system $M/G/1/FCFS/\infty/\infty$.

Since the service times may be nonexponential, the probability that a death (service completion) occurs between time t and $t + \Delta t$ will depend on the time since the last service completion.

This violates **Law 2**, so we cannot model an $M/G/1/FCFS/\infty/\infty$ system as a birth-death process.

Birth & Death Process Transitions (p. 1066)

$P_{ij}(t)$ = probability that the queuing system will be in state j at time $t + \Delta t$ given that it is in state i at time 0

Case	State at time t	State at time $t + \Delta t$	Probability
(I)	$j - 1$	j	$P_{i,j-1}(t)[\lambda_{j-1}\Delta t + o(\Delta t)]$
(II)	$j + 1$	j	$P_{i,j+1}(t)[\mu_{j+1}\Delta t + o(\Delta t)]$
(III)	j	j	$P_{i,j}(t)[1 - \mu_j\Delta t - \lambda_j\Delta t - 2o(\Delta t)]$
(IV)	other	j	$o(\Delta t)$

Therefore,

$$\begin{aligned}P_{i,j}(t + \Delta t) &= (I) + (II) + (III) + (IV) \\&= \Delta t[\lambda_{j-1}P_{i,j-1}(t) + \mu_{j+1}P_{i,j+1}(t) - \mu_jP_{i,j}(t) - \lambda_jP_{i,j}(t)] \\&\quad + P_{i,j}(t) + o(\Delta t)\end{aligned}$$

Birth & Death Process Transitions (p. 1066)

$$P_{i,j}(t + \Delta t) = \Delta t[\lambda_{j-1}P_{i,j-1}(t) + \mu_{j+1}P_{i,j+1}(t) - \mu_j P_{i,j}(t) - \lambda_j P_{i,j}(t)] + P_{i,j}(t) + o(\Delta t)$$

Now let

$$\lambda_j = \lambda, \quad j = 0, 1, 2, \dots$$

$$\mu_j = \mu, \quad j = 1, 2, 3, \dots$$

$$\mu_0 = 0$$

Then, for $j \geq 1$:

$$\lim_{\Delta t \rightarrow 0} \frac{P_{i,j}(t + \Delta t) - P_{i,j}(t)}{\Delta t} = \lim_{\Delta t \rightarrow 0} [\lambda P_{i,j-1}(t) + \mu P_{i,j+1}(t) - \mu P_{i,j}(t) - \lambda P_{i,j}(t)] + \lim_{\Delta t \rightarrow 0} \frac{o(\Delta t)}{\Delta t}$$



$$P'_{i,j}(t) = \lambda P_{i,j-1}(t) + \mu P_{i,j+1}(t) - \mu P_{i,j}(t) - \lambda P_{i,j}(t)$$

Birth & Death Process Transitions (p. 1066)

$$P_{i,j}(t + \Delta t) = \Delta t[\lambda_{j-1}P_{i,j-1}(t) + \mu_{j+1}P_{i,j+1}(t) - \mu_j P_{i,j}(t) - \lambda_j P_{i,j}(t)] + P_{i,j}(t) + o(\Delta t)$$

Now let

$$\lambda_j = \lambda, \quad j = 0, 1, 2, \dots$$

$$\mu_j = \mu, \quad j = 1, 2, 3, \dots$$

$$\mu_0 = 0$$

Then, for $j = 0$:

$$\lim_{\Delta t \rightarrow 0} \frac{P_{i,0}(t + \Delta t) - P_{i,0}(t)}{\Delta t} = \lim_{\Delta t \rightarrow 0} [\mu P_{i,1}(t) - \lambda P_{i,0}(t)] + \lim_{\Delta t \rightarrow 0} \frac{o(\Delta t)}{\Delta t}$$



$$P'_{i,0}(t) = \mu P_{i,1}(t) - \lambda P_{i,0}(t)$$

Birth & Death Process Steady States (pp. 1067–1069)

We therefore have the system of differential equations:

For $j \geq 1$: $P'_{i,j}(t) = \lambda P_{i,j-1}(t) + \mu P_{i,j+1}(t) - \mu P_{i,j}(t) - \lambda P_{i,j}(t)$

For $j = 0$: $P'_{i,0}(t) = \mu P_{i,1}(t) - \lambda P_{i,0}(t)$

Letting $P_{i,j}(t) = \pi_j$ in the steady state as $t \rightarrow \infty$ and noting that $\pi'_j = 0$, this system of differential equations reduces to the system of algebraic equations:

$$\begin{aligned}\lambda\pi_{j-1} + \mu\pi_{j+1} - \mu\pi_j - \lambda\pi_j &= 0, & j = 1, 2, 3, \dots \\ \mu\pi_1 - \lambda\pi_0 &= 0,\end{aligned}$$

which has a solution

$$\pi_j = \rho^j(1 - \rho) \text{ for all } j = 0, 1, 2, \dots$$

where $\rho = \frac{\lambda}{\mu}$.